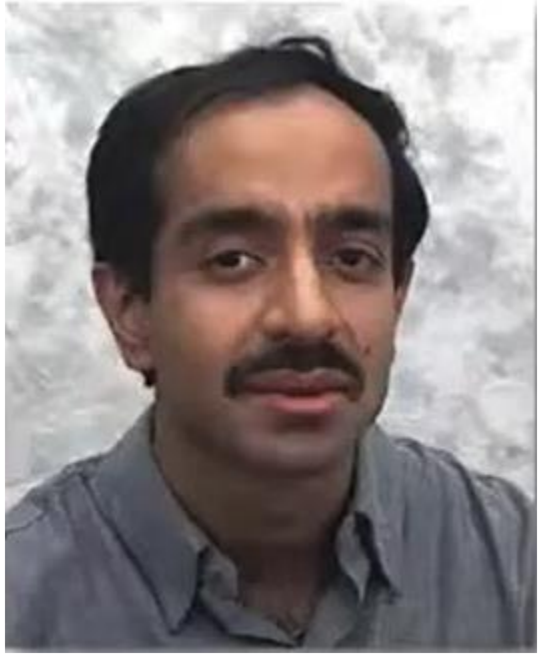
Quantum Database Search: Grover's Algorithm

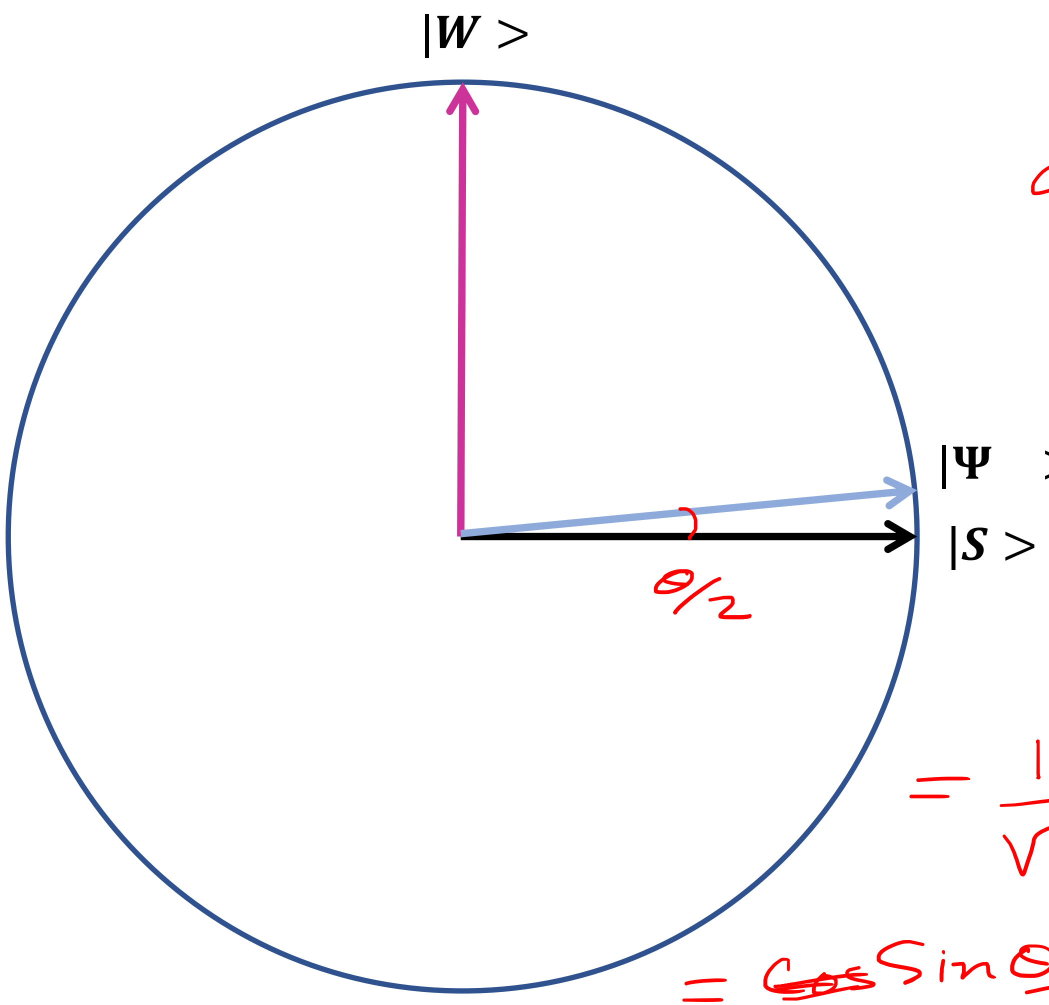
Finding a needle in a haystack



Lov K. Grover

Amlan Mukherjee

Grover's algo: visualized



$$\alpha^2 + \beta^2 = 1$$

$$|\Psi\rangle = \sum \frac{1}{\sqrt{N}} |x\rangle$$

$$= \alpha |w\rangle + \beta |w_\perp\rangle$$

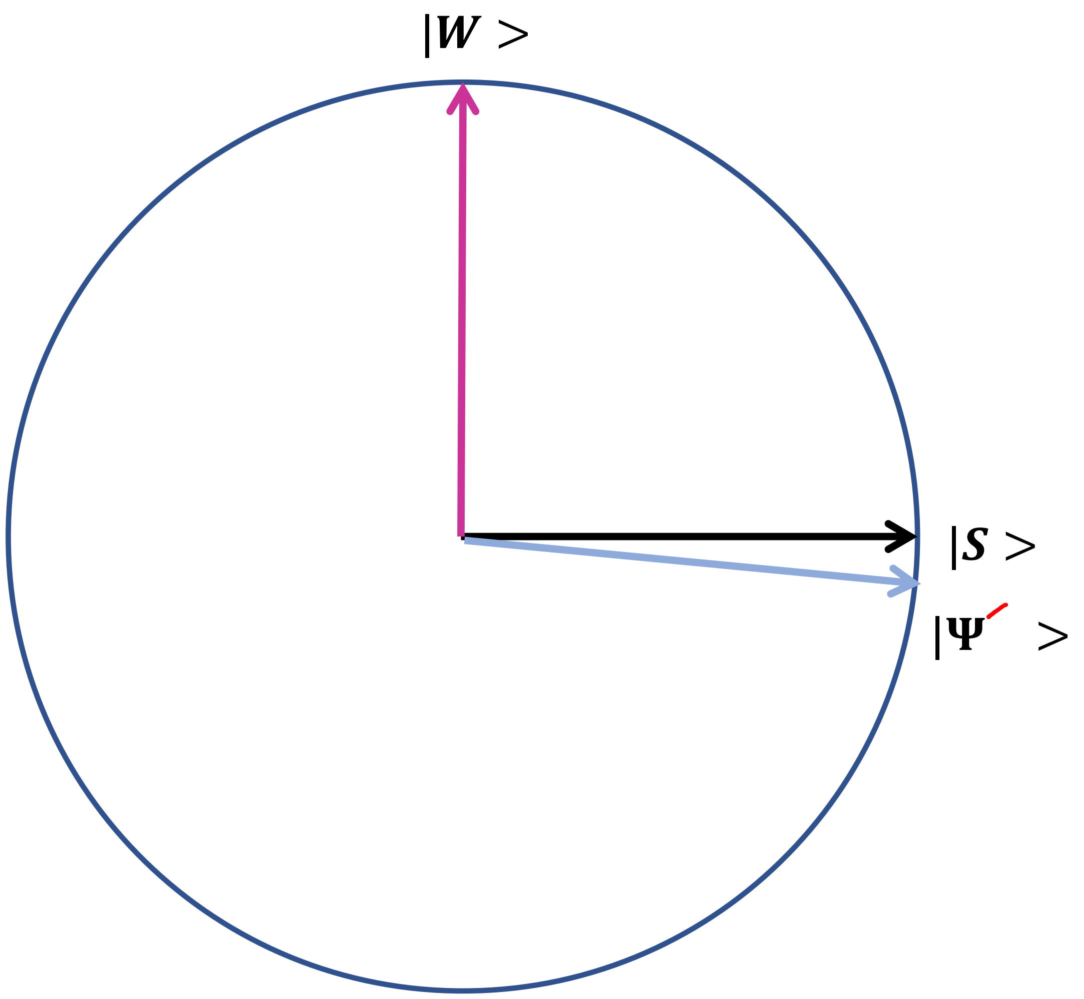
$$= \frac{1}{\sqrt{N}} |w\rangle + \frac{\sqrt{N-1}}{\sqrt{N}} |w_\perp\rangle$$

$$= \cancel{\cos} \sin\frac{\theta}{2} |w\rangle + \cos\theta/2 \, |\cancel{w_\perp}\rangle \; |s\rangle$$

$$\sin\theta/2 = \frac{1}{\sqrt{N}}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

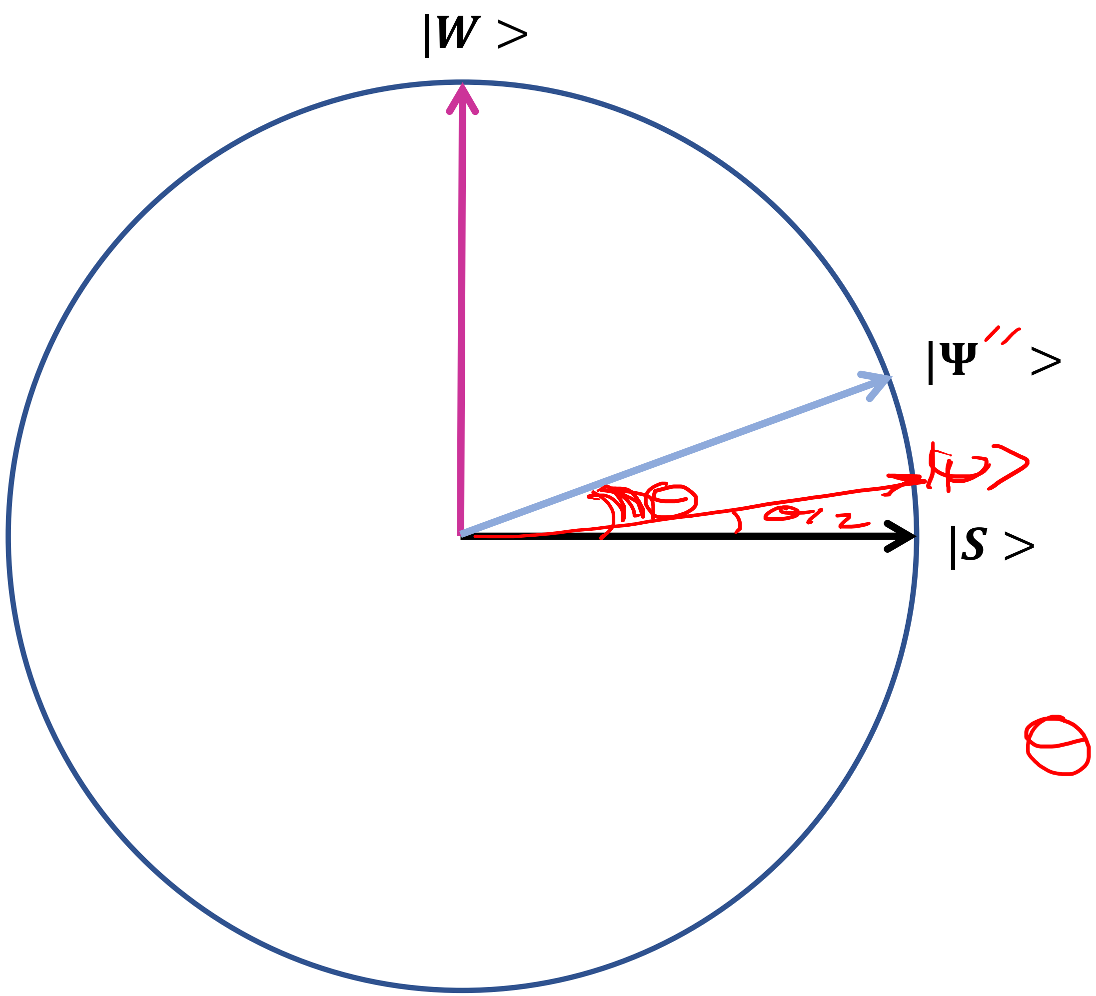
Grover's algo: visualized



① Phase Inv.

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: visualized

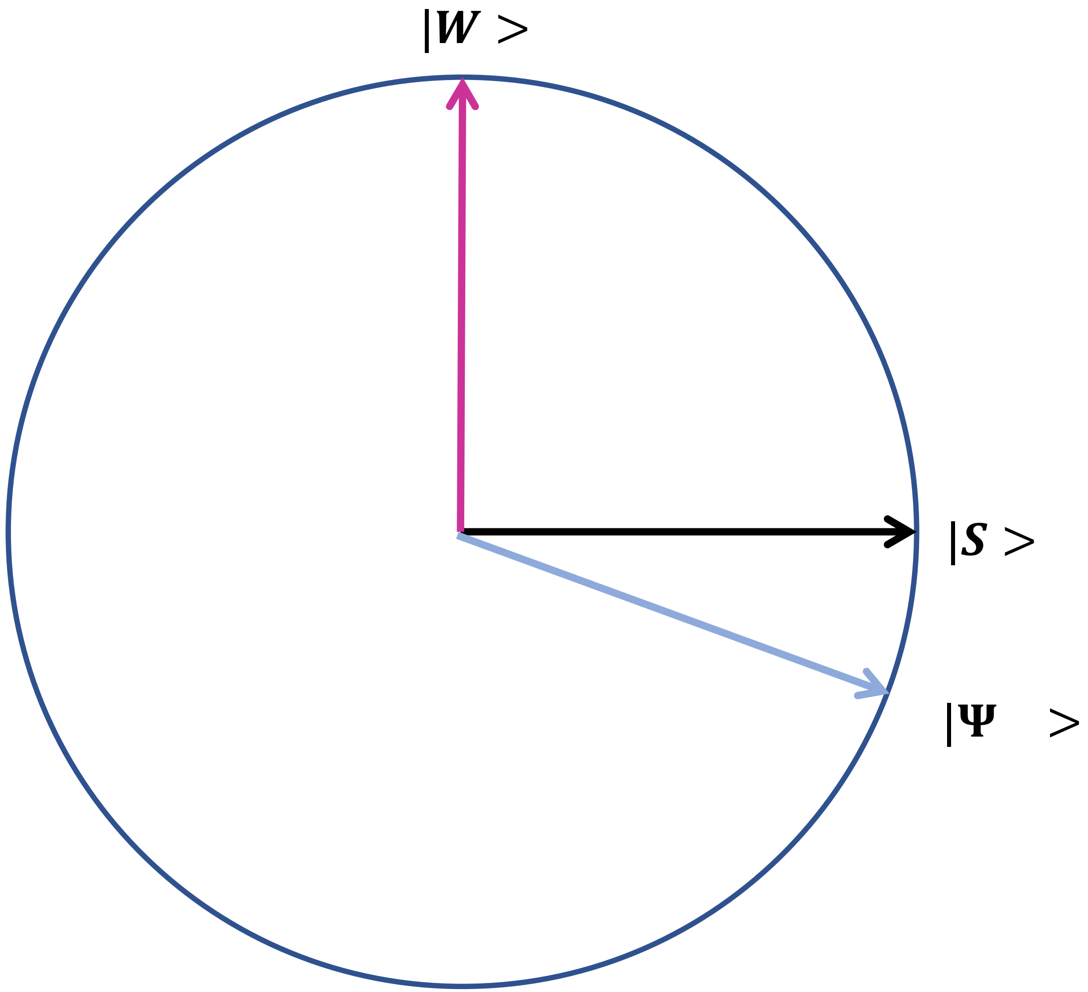


② Inv over
mean

$\theta + \theta/2$

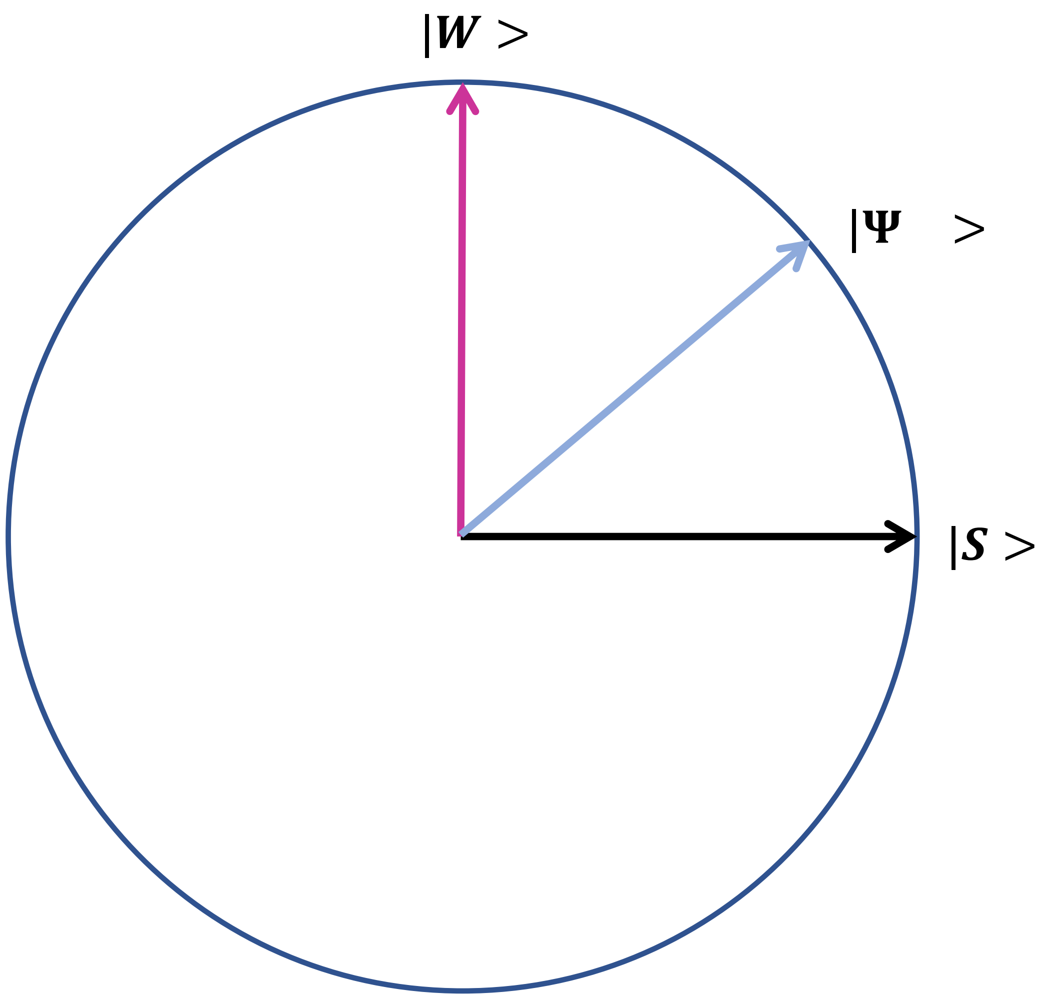
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover’s algo: visualized



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: visualized



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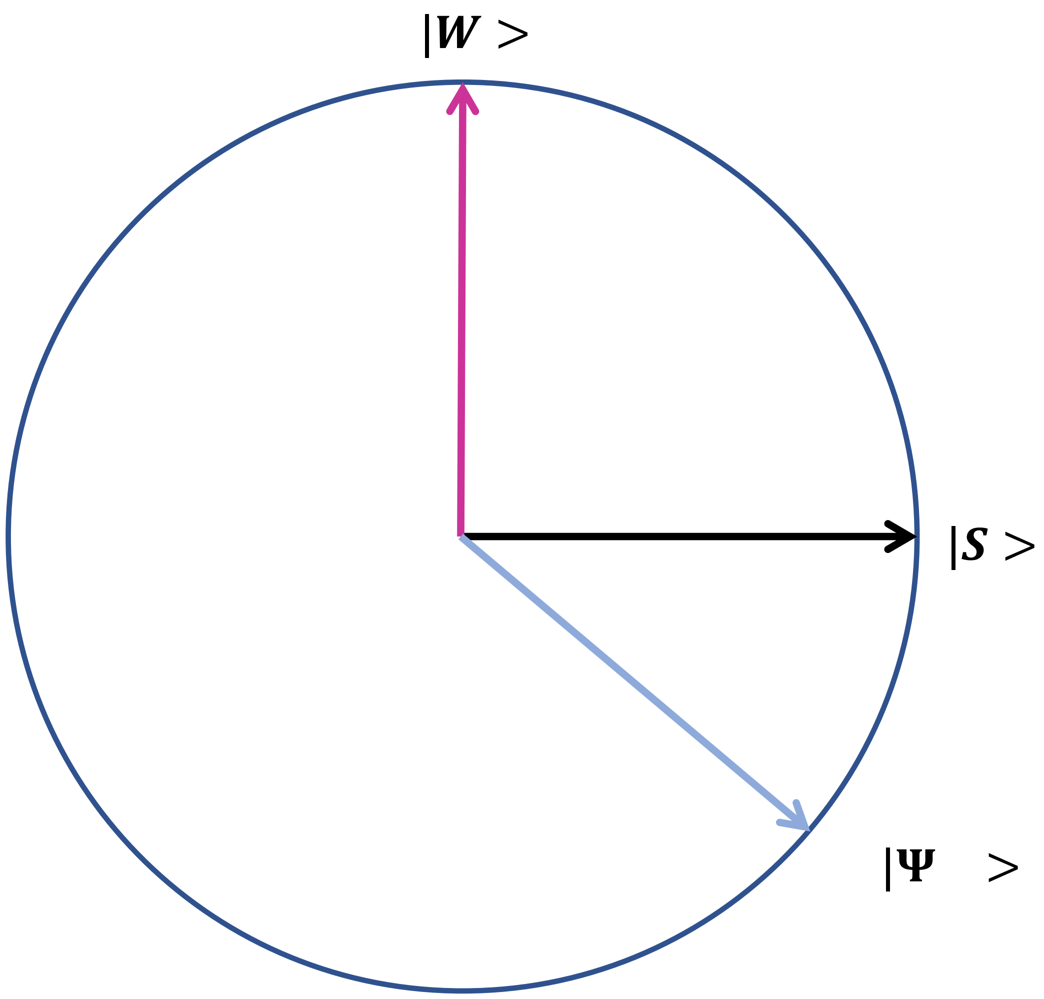
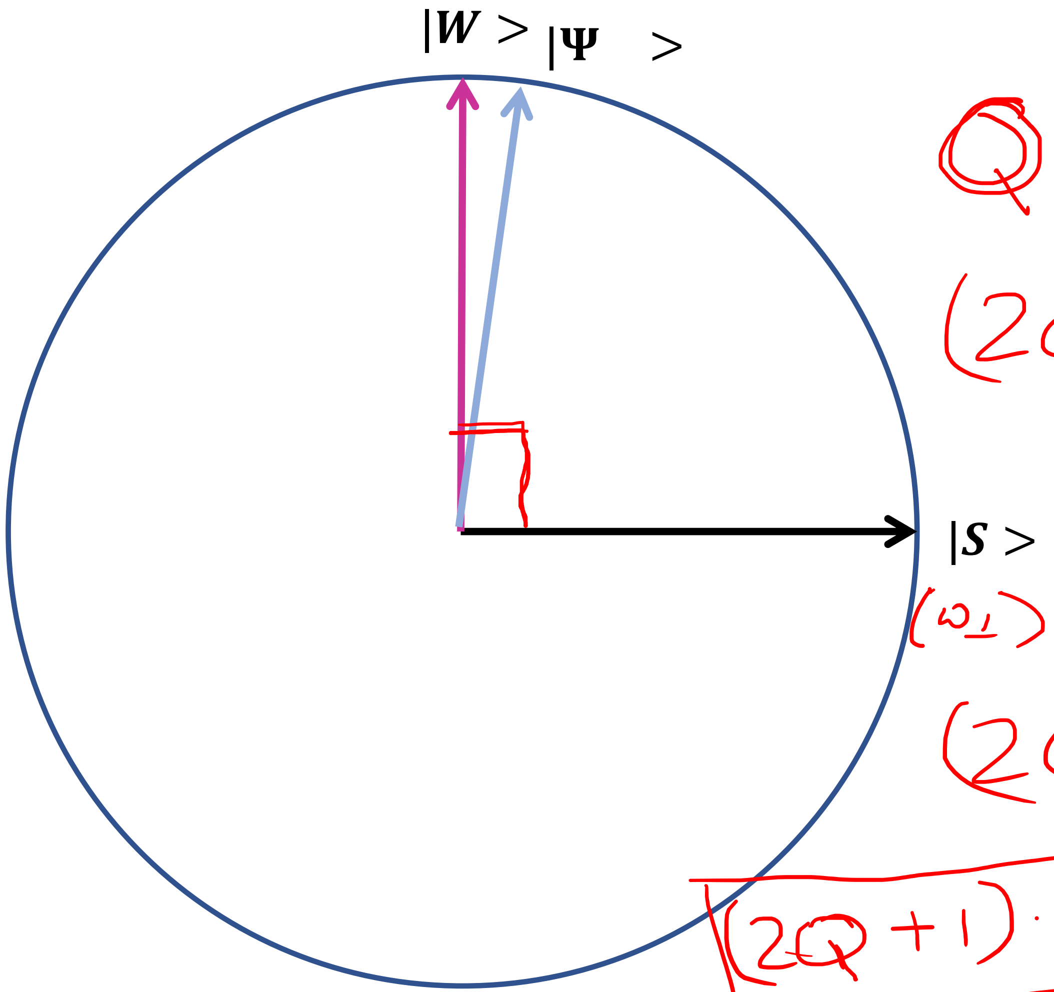
Grover's algo: visualized



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: visualized



$$Q \cdot \theta + \theta/2 = \pi/2$$

$$(2Q+1)\theta = \pi$$

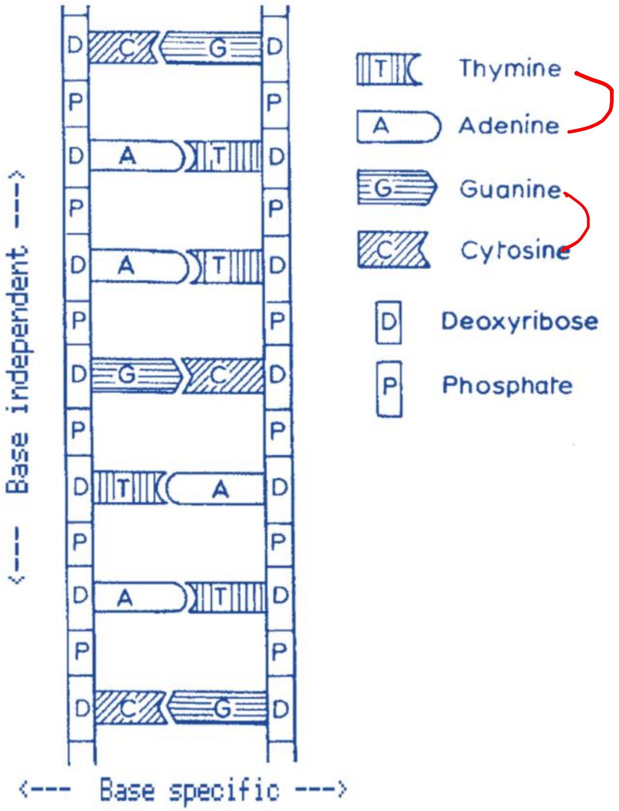
$$\sin \theta/2 = \frac{1}{\sqrt{N}}$$

$$(2Q+1)\, 2\sin^{-1}\left(\frac{1}{\sqrt{N}}\right) = \pi$$

$$\boxed{(2Q+1) \cdot \sin^{-1}\left(\frac{1}{\sqrt{N}}\right) = \frac{\pi}{2}}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Applications: DNA



$$\boxed{(2Q + 1)\, sin^{-1}(1/\sqrt{N}) = \pi/2} \quad 1$$

$Q = 1; N = 4$ $\quad Q = 2; N = 10.5$ $\quad Q = 3; N = 20.2$

"A single base-pairing distinguishes between 4 possibilities in DNA replication and mRNA transcription. This is an exact solution of eq. (1), so chances of error are minimized"

" N = 10 is not an exact solution for Q = 2, which means that the quantum algorithm will not always find the desired object. There exists a small probability, about 1 part in 1000, that the quantum algorithm will select a wrong object."

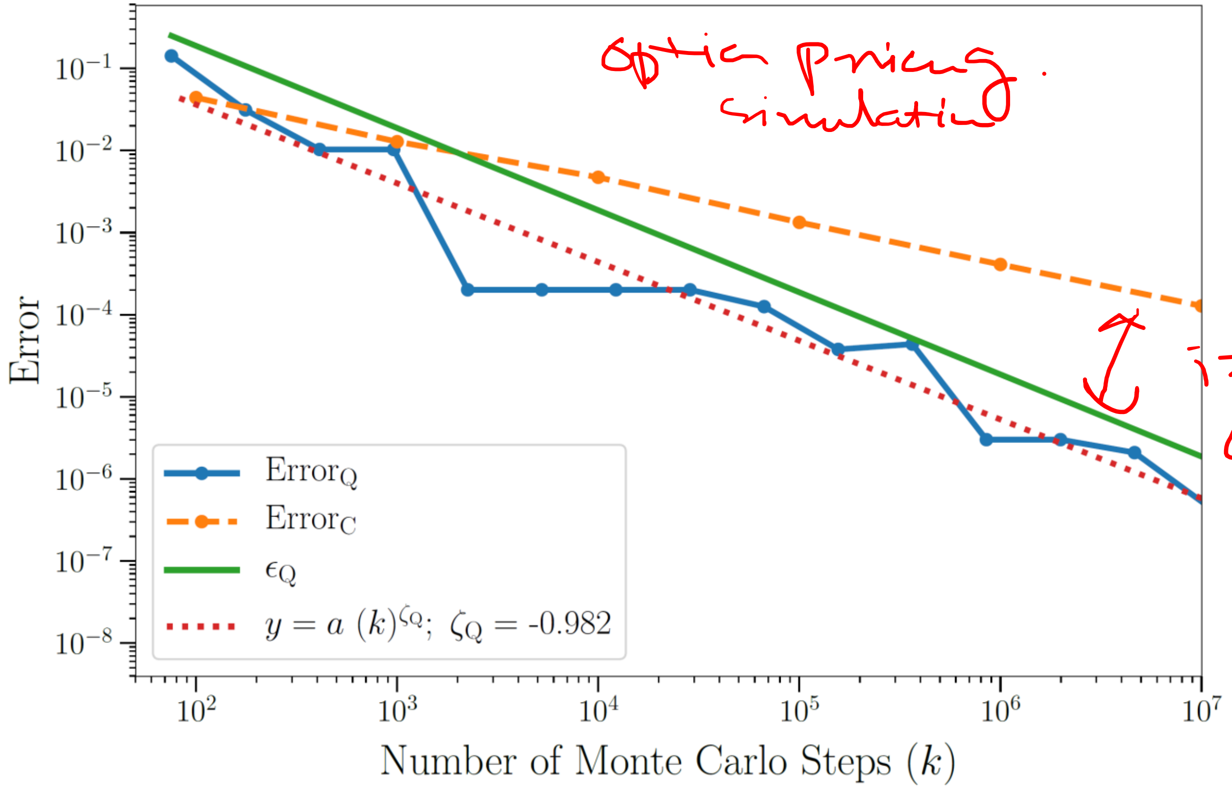
Ref: Apoorva Patel, J. Biosci. 26 (2), 145 (2001)

Ref: Apoorva Patel, arXiv preprint quant-ph/0002037 (2000)

Applications: Many more

- Function optimization
- Markov chain algorithms
- Montecarlo simulation

- Traveling salesman problem (TSP).
- Graph 3-coloring.
- Image pattern matching



option pricing simulation.

improvement over classical

Ref: arXiv:1805.00109v2 [quant-ph] (2018)

Ref: arXiv:quant-ph/0405001 (2004)

Ref: arXiv:2108.10854v2 [quant-ph] (2021)

Thank you for your attention!